

Youssef Alaa Alfahl

Communication and Information Engineer | Data Science | Full-Stack Development

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Professional Summary

Communication and Information Engineering graduate with a strong academic record and hands-on experience in machine learning, deep learning, data analysis, and full-stack development. Built academic and practical projects across software engineering, mobile/web development, digital communications, and statistical analysis. Seeking opportunities in software engineering, data science, machine learning, or applied engineering roles.

Experience

University of Science and Technology in Zewail City

[01/07/2024 – 30/9/2024]

Sector: Communication and Information Engineering | **Role:** Research Assistant

Project: Deep Learning-Based RF Wave Modeling and Power Distribution Using Ray Tracing in Indoor Environments.

- **Simulation:** Generated indoor RF power maps in MATLAB using 3D modeling and ray tracing showing antenna patterns, reflections, materials.
- **Modeling:** Built power maps at user height and validated a U-Net to predict maps.
- **Outcome:** End-to-end pipeline for fast indoor coverage estimation across multiple scenarios.

National University of Sciences and Technology (NUST), Islamabad, Pakistan

[08/07/2025 – 14/08/2025]

Sector: School of Electrical Engineering and Computer Science (SEECs) | **Role:** Research Intern

Project: Multitask Deep Learning for Real-Time Optimization of Polarization Snapshots in Rotating-Polarization BPSK/QPSK

- **Simulation:** Modeled RP BPSK/QPSK links (AWGN/Rayleigh/Rician) and generated a Monte Carlo dataset in MATLAB.
- **Modeling:** Labeled optimal N_p under latency/ops-per-bit constraints and trained a multitask DL model to select N_p and predict BER.
- **Outcome:** Closely matched brute-force decisions with much lower computation, enabling real-time adaptive RP link configuration.

DHCB, Giza, Egypt

[01/10/2025 – 1/12/2025]

Sector: Software Development | **Role:** Full Stack Developer

Built and maintained frontend interfaces, backend APIs, and database-integrated features for software applications in a collaborative development environment.

University of Science and Technology in Zewail City

[01/04/2026 – Present]

Sector: Education | **Role:** Part-Time Teaching Assistant | **Course:** CSCI 101 – Introduction to Computer Science (Python)

Support undergraduate students by teaching Python programming and guiding them through core problem-solving concepts.

Education

B.S.C., Communication and Information Engineering, University of Science and Technology in Zewail City

Enrollment: 03/10/2021 | **Graduation:** 01/02/2026 | **Final grade:** 3.80

Projects

Memento, Context-Aware Daily Task & Memory Assistant | Graduation Project

[01/11/2024 – 01/06/2025]

- **Mobile app (Flutter):** Built a mobile assistant with passive/active modes, speech capture (VAD), and natural querying.
- **Backend & intelligence (Python):** Integrated Gemini 2.5 Flash for event understanding/summarization, with AI-agent orchestration and WebSockets for real-time interaction.
- **Core features:** Offline-first storage and sync, GPS/time-based reminders through notifications, and full event CRUD/search with conflict detection & disambiguation.

Friendza, Social Media Mobile Application

- **Mobile app (React Native CLI):** Built a social media app with posts, stories, profiles, likes, comments, saves, and follow interactions using React Navigation for a smooth mobile experience.
- **Backend & database (Node.js, Express, PostgreSQL):** Developed a RESTful backend to handle authentication, user profiles, media posts, stories, and social interactions.
- **Core features:** Added light/dark themes, English/Arabic language support with RTL layout, privacy controls, camera/gallery uploads, video playback, and persistent app settings.

PlacePulse, Community App for Sharing Place Experiences

- **Web app (React):** Built a community-driven platform where users can discover, share, review, and save places, with profiles, collections, media uploads, and responsive UI components.
- **Backend & database (Node.js, Express, Mongoose):** Developed a full-stack MERN application with JWT authentication, place and review management, collection sharing, validation, and data integrity handling.
- **AI & location features:** Integrated Gemini-powered review summaries, place Q&A, and semantic search, alongside geocoding and interactive maps for richer place exploration.

COVID-19 Case Surveillance (CDC), Statistical Inference Project

- **Data analysis pipeline (Python, DuckDB, Parquet):** Built a scalable workflow to process and analyze a large CDC COVID-19 surveillance dataset with approximately **19 million rows**, including validation, preprocessing, and optimized storage.
- **Statistical analysis:** Performed descriptive analytics, hypothesis testing, and logistic regression to study associations between demographics, symptoms, risk factors, and outcomes.
- **Reporting & insights:** Produced visualizations and report-ready tables highlighting trends, subgroup patterns, and outcome differences while accounting for missing and incomplete data.

Embedded Weather Monitoring & IoT Dashboard (TM4C123 + ESP8266 + Firebase + Flutter | FreeRTOS)

- **Embedded system (TM4C123 + FreeRTOS):** Built a sensing node to measure temperature, humidity, and light intensity using task/queue-based data collection and UART communication.
- **IoT gateway & cloud (ESP8266, Firebase):** Developed a gateway to display readings on an I²C LCD and upload sensor data to Firebase Realtime Database over HTTPS.
- **Mobile app (Flutter):** Created a monitoring app to display live readings, history, and trend graphs, with anomaly detection and push notifications.

OFDM Wireless Link Simulator (IEEE 802.11a-Style) | Python

- **PHY simulation (Python):** Built an end-to-end OFDM wireless link simulator from MP3 input to recovered audio, including coding, interleaving, modulation, subcarrier mapping, IFFT, and cyclic prefix.
- **Channel & receiver modeling:** Simulated realistic transmitter, channel, and receiver effects including PAPR reduction, amplifier nonlinearity, multipath, AWGN, channel estimation, and equalization.
- **Performance evaluation:** Assessed link quality using BER vs SNR, EVM, constellation plots, PSD/spectrogram, and PAPR CCDF metrics.

Technical Skills

Programming: Python, C/C++, JavaScript, SQL, MATLAB

Web & Mobile: React, React Native, Next.js, Node.js, Express.js, MERN Stack, Flutter, REST APIs

Data & ML: Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Seaborn, DuckDB

Databases & Cloud: PostgreSQL, MongoDB, Mongoose, Firebase

Tools: Git, GitHub, SystemVerilog, ModelSim, LTspice, CST Studio Suite, LabVIEW

Activities

- **Zewail City Science Festival 2022:** I contributed as a content creator and actor in scientific mini plays.
- **IEEE Zewail City Club:** I designed and conducted scientific experiments to illustrate key scientific concepts in different fields.